

Video Presentation

Unsupervised Machine Learning for Customer Churn Analysis in Telecom Companies



CSCA 5623 Unsupervised Machine Learning Final Project
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AGENDA ITEMS

1. Problem Description
2. Data and Exploratory Data Analysis
3. Building Unsupervised Models
4. Demo
5. Results and Analysis
6. Conclusions



1. Project Problem

Project Topic:

Customer churn analysis in telecom companies traditionally relies on supervised learning to predict which customers are likely to leave. However, **unsupervised machine learning offers an alternative approach by uncovering hidden patterns and segmenting customers without predefined labels.** This project explores how unsupervised learning techniques can help telecom companies identify at-risk customer groups, detect anomalies, and derive actionable insights for retention strategies.

1. Project Problem

Why Unsupervised Learning for Churn Analysis?

1. **No Need for Labeled Data** – Unlike supervised models, unsupervised learning does not require historical churn labels, making it useful when labeled data is scarce or unreliable.
2. **Discover Hidden Patterns** – Clustering and anomaly detection can reveal unexpected customer behaviors that may lead to churn, beyond what traditional models capture.
3. **Customer Segmentation** – Grouping customers based on behavior, usage, and demographics helps tailor retention strategies for different segments.
4. **Early Anomaly Detection** – Unsupervised models can detect unusual activity (e.g., sudden drops in usage) that may indicate potential churn before it happens.

1. Project Problem

Key Objectives:

- **Apply Clustering Algorithms** (e.g., K-Means, Hierarchical Clustering) to group customers with similar behavior.
- **Detect Anomalies** using techniques like Isolation Forest or Autoencoders to flag unusual customer activity.
- **Analyze Feature Importance** to understand which factors contribute most to customer groupings.
- **Develop Retention Strategies** based on unsupervised insights, such as personalized offers for high-risk clusters.

Expected Outcomes:

- Identification of distinct **customer segments** with varying churn risks.
- Detection of early warning signs of churn through **anomaly detection**.
- **Data-driven recommendations** for targeted customer retention campaigns.

2. Data

Data Source Description:

Context Data Source:

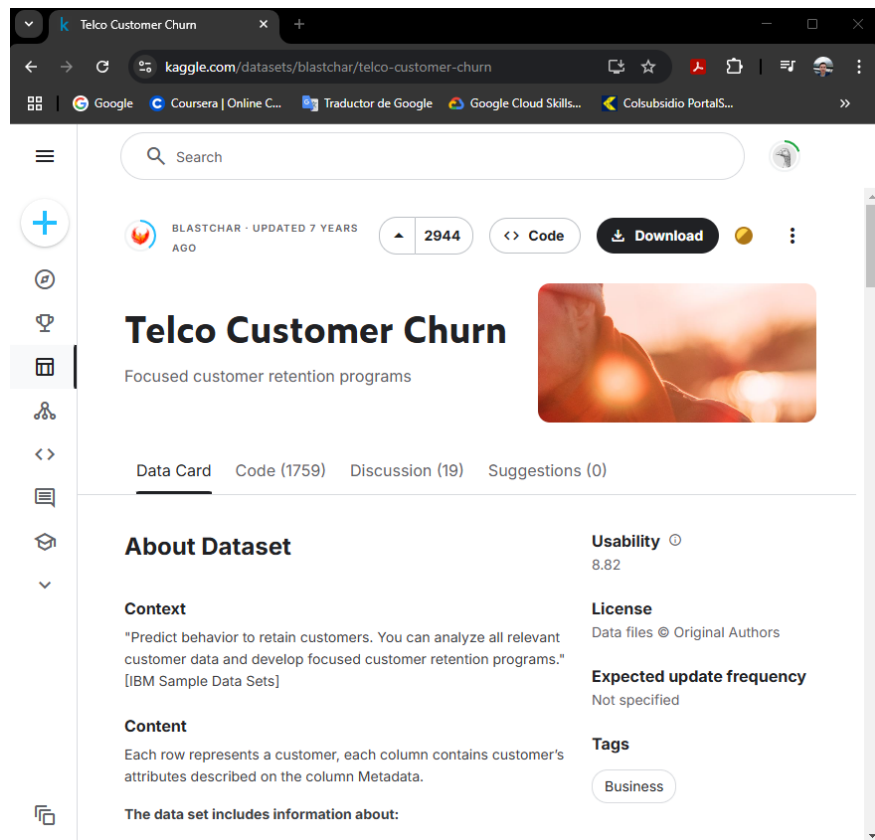
- **Dataset Name:** Telco Customer Churn (focused customer retention programs)
- **Url:** <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
- **Motivation:** "Predict behavior to retain customers. You can analyze all relevant customer data and develop focused customer retention programs." [IBM Sample Data Sets]

Content:

Each row represents a customer, each column contains customer's attributes described on the column Metadata.

The data set includes information about:

- Customers who left within the last month – the column is called Churn
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges
- Demographic info about customers – gender, age range, and if they have partners and dependents



The screenshot shows the Kaggle dataset page for 'Telco Customer Churn' by Blastchar. The page is viewed in a browser with the URL 'kaggle.com/datasets/blastchar/telco-customer-churn'. The dataset has 2944 rows and was updated 7 years ago. It includes buttons for 'Code', 'Download', and a 'Usability' score of 8.82. The 'About Dataset' section contains the same motivation and content descriptions as the text on the left. The 'Expected update frequency' is 'Not specified', and the 'License' is 'Data files © Original Authors'. A 'Tags' section shows 'Business' as a tag.

Telco Customer Churn
Focused customer retention programs

About Dataset

Context
"Predict behavior to retain customers. You can analyze all relevant customer data and develop focused customer retention programs." [IBM Sample Data Sets]

Content
Each row represents a customer, each column contains customer's attributes described on the column Metadata.

The data set includes information about:

Usability 8.82

License
Data files © Original Authors

Expected update frequency
Not specified

Tags
Business

2. Data

Dataset Variables

#	VARIABLES	#	VARIABLES
1	customerID: Customer ID	11	OnlineBackup: Whether the customer has online backup or not (Yes, No, No internet service)
2	gender: Whether the customer is a male or a female	12	DeviceProtection: Whether the customer has device protection or not (Yes, No, No internet service)
3	SeniorCitizen: Whether the customer is a senior citizen or not (1, 0)	13	TechSupport: Whether the customer has tech support or not (Yes, No, No internet service)
4	Partner: Whether the customer has a partner or not (Yes, No)	14	StreamingTV: Whether the customer has streaming TV or not (Yes, No, No internet service)
5	Dependents: Whether the customer has dependents or not (Yes, No)	15	StreamingMovies: Whether the customer has streaming movies or not (Yes, No, No internet service)
6	tenure: Number of months the customer has stayed with the company	16	Contract: The contract term of the customer (Month-to-month, One year, Two year)
7	PhoneService: Whether the customer has a phone service or not (Yes, No)	17	PaperlessBilling: Whether the customer has paperless billing or not (Yes, No)
8	MultipleLines: Whether the customer has multiple lines or not (Yes, No, No phone service)	18	PaymentMethod: The customer's payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic))
9	InternetService: Customer's internet service provider (DSL, Fiber optic, No)	19	MonthlyCharges: The amount charged to the customer monthly
10	OnlineSecurity: Whether the customer has online security or not (Yes, No, No internet service)	20	TotalCharges: The total amount charged to the customer
		21	Churn: Whether the customer churned or not (Yes or No)

3. Data Cleaning

Data Overview:

- Reading the dataset
- Understanding the shape of the dataset
- Checking the data types
- Checking for missing values
- Checking for duplicated values
- Identification of categorical and numerical features

```
df=pd.read_csv("./data/telco-customer-churn.csv")
```

```
# Head info
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService
0	7590-VHVEG	Female	0	Yes	No	1	No
1	5575-GNVDE	Male	0	No	No	34	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes
3	7795-CFOCW	Male	0	No	No	45	No
4	9237-HQITU	Female	0	No	No	2	Yes

5 rows × 21 columns

```
# Head info
print(df.shape)
```

(7843, 21)

3. Data Cleaning:

```
# 1. Drop 'customerID' as this column does not gives important information
# in the data set
df.drop('customerID', axis=1, inplace=True)

# 2. Convert Yes o No to binary (1,0)
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})

# 3. Data cleaning for the column 'TotalCharges'
blank_indexes = df[df['TotalCharges'] == ''].index.values.tolist()
#blank_indexes , _ = df[df['TotalCharges'] == ''].index
#print(blank_indexes)

# As we found blank string values for some 'TotalCharges' it will be useful
# to assign the value as 'MonthlyCharges' exist
for index in blank_indexes:
    df.loc[index, 'TotalCharges'] = df.loc[index, 'MonthlyCharges']
    #print(df.iloc[index])

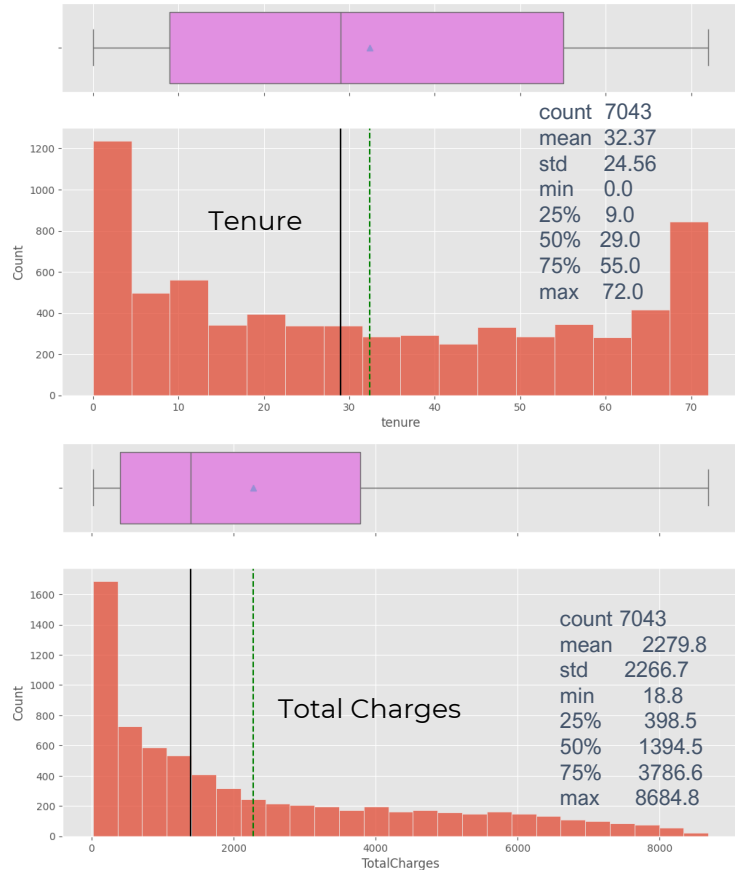
#Convert TotalCharges to numerical
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"])
```

Setting categorical and numerical variables

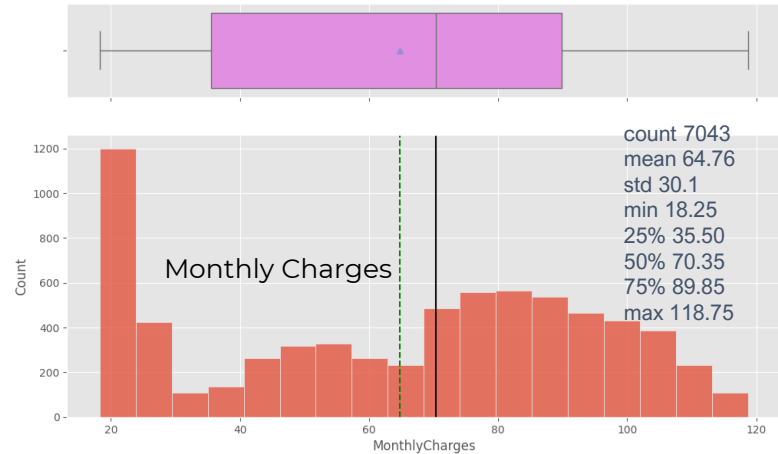
```
# Categorical variables
categorical_cols = ['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines',
'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod']

# Numerical variables
numerical_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']
```


2. Exploratory Data Analysis

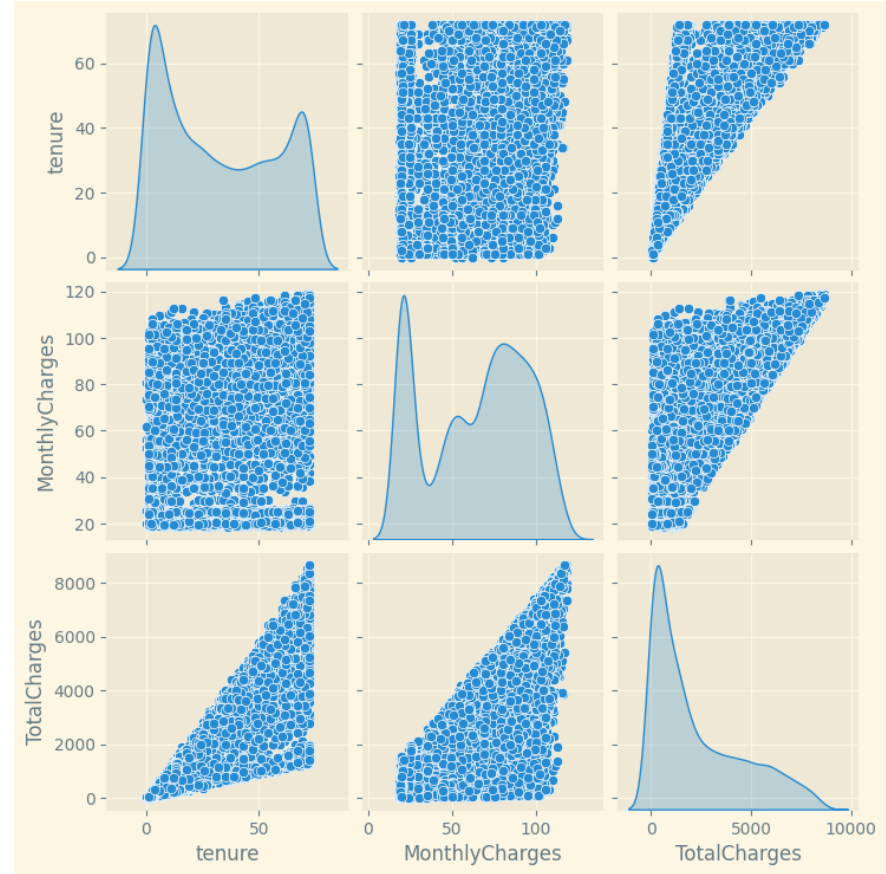
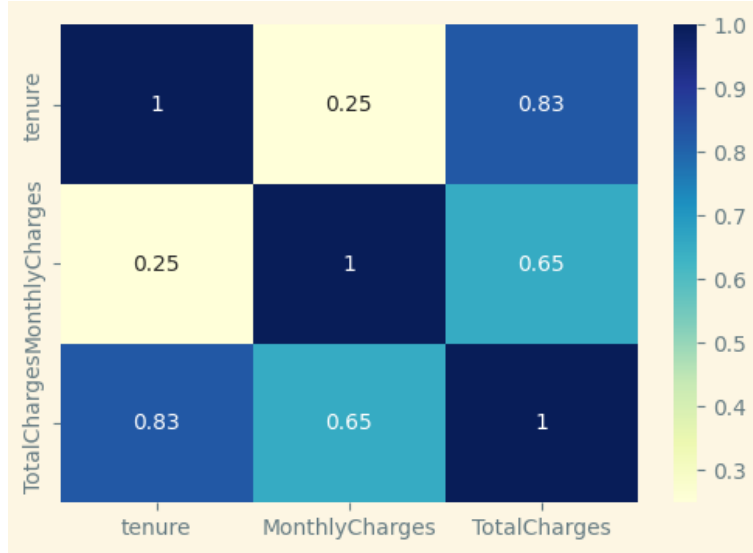


Boxplot and histogram combined analysis



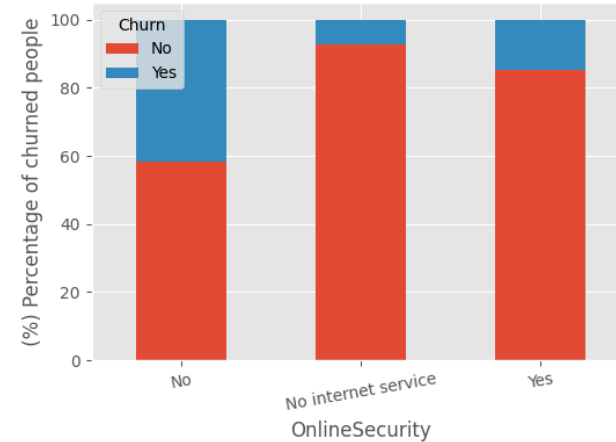
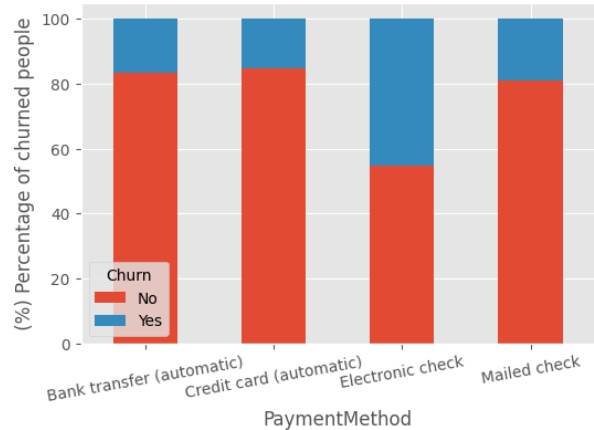
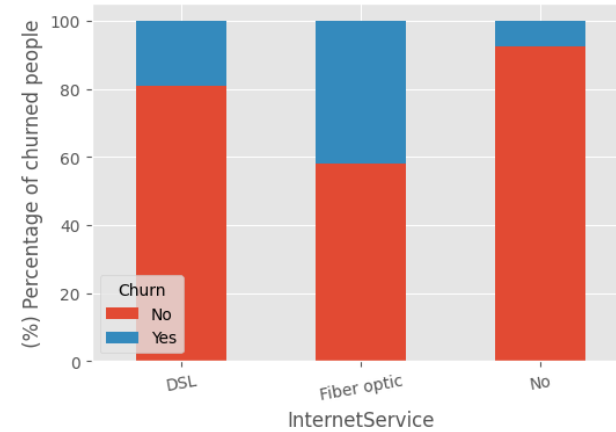
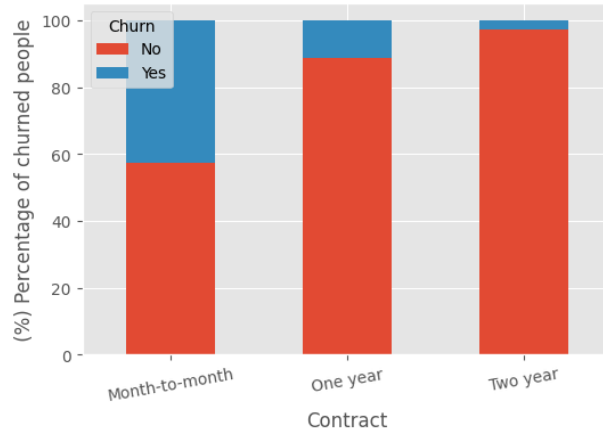
2. Exploratory Data Analysis

Multivariate Analysis



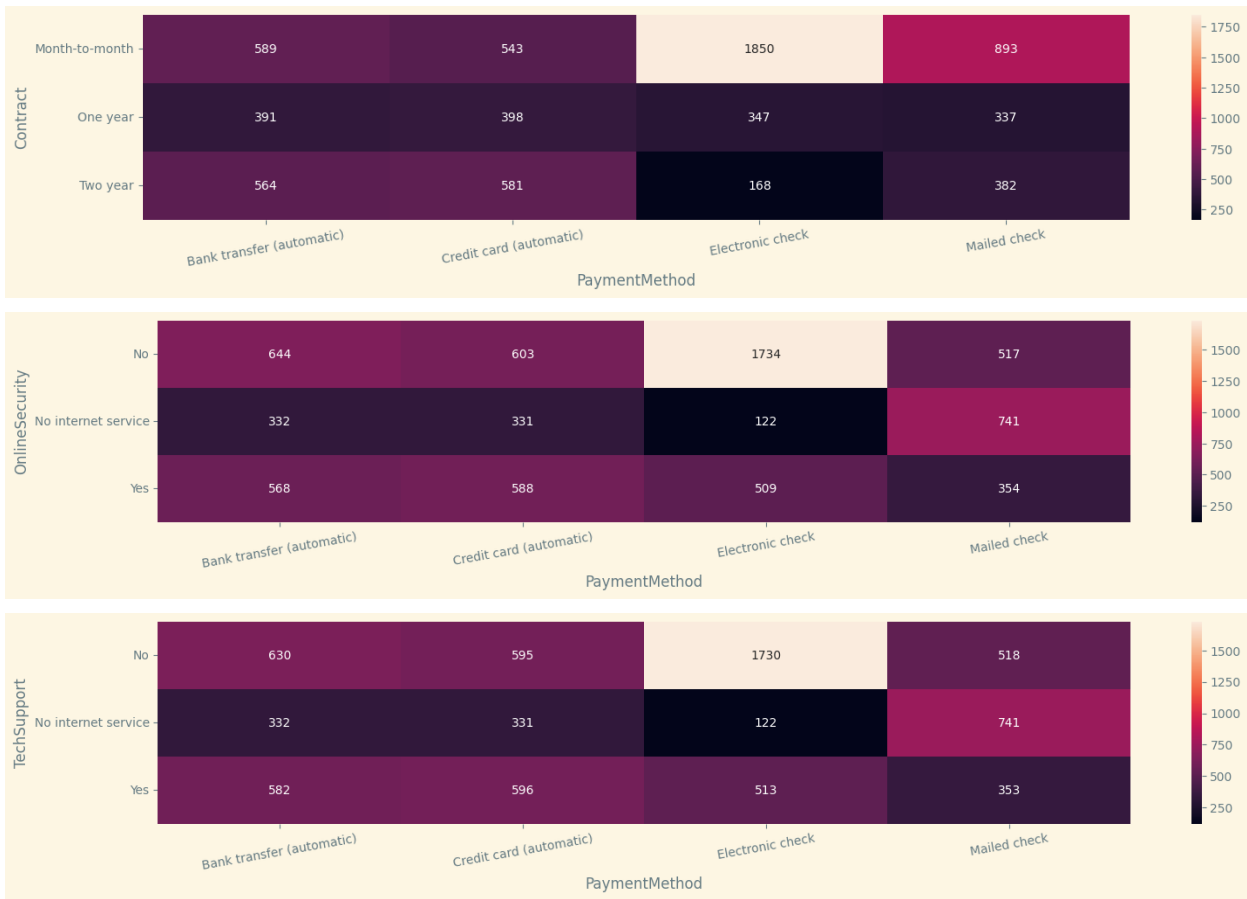
2. Exploratory Data Analysis

Bivariate Analysis

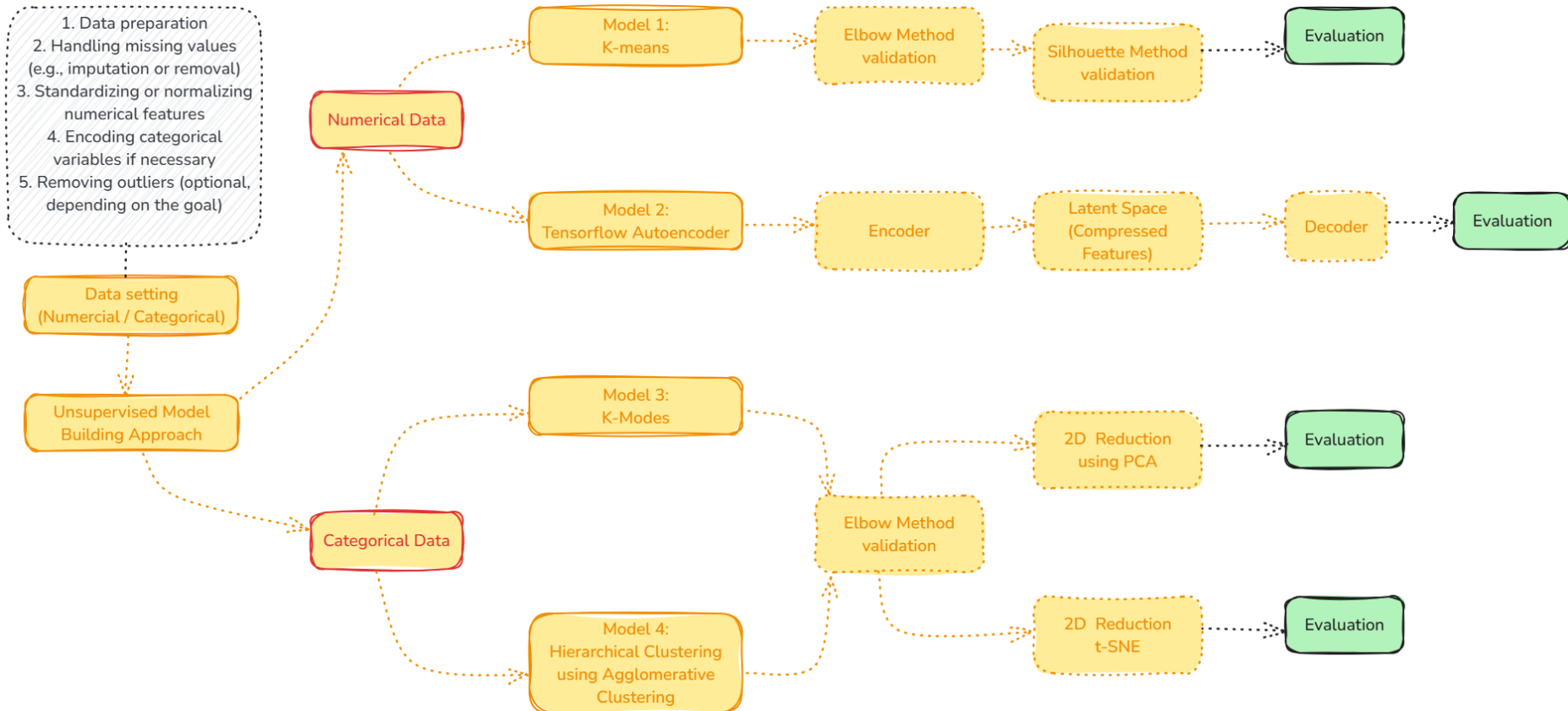


4. Exploratory Data Analysis

Review of the relationships of categorical data



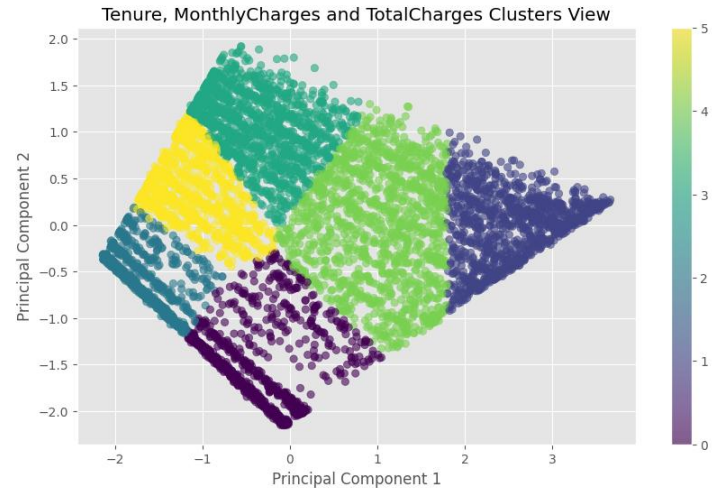
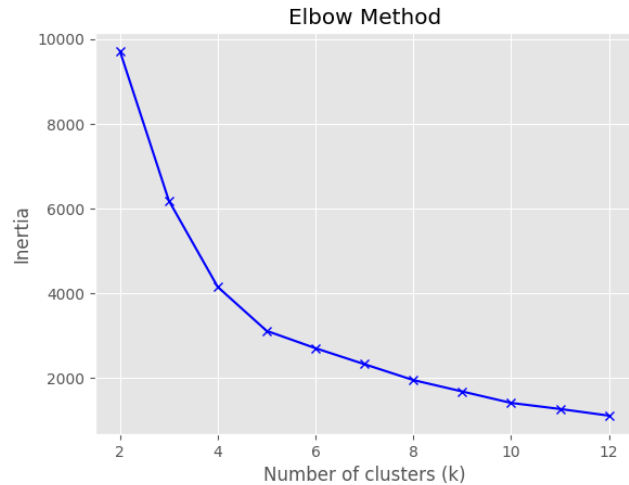
3. Building Unsupervised Models



5. Demo



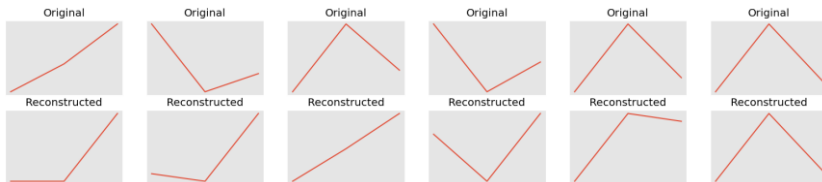
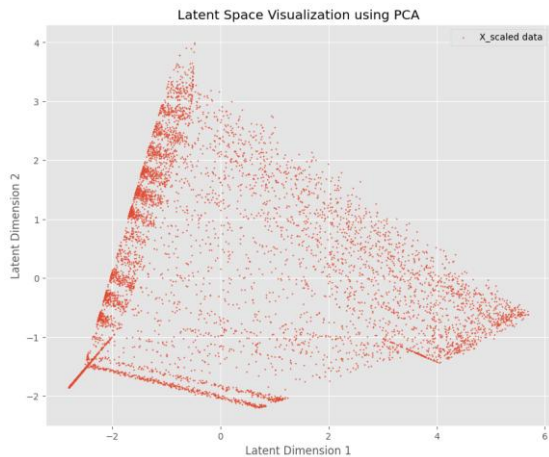
5. Results and Analysis



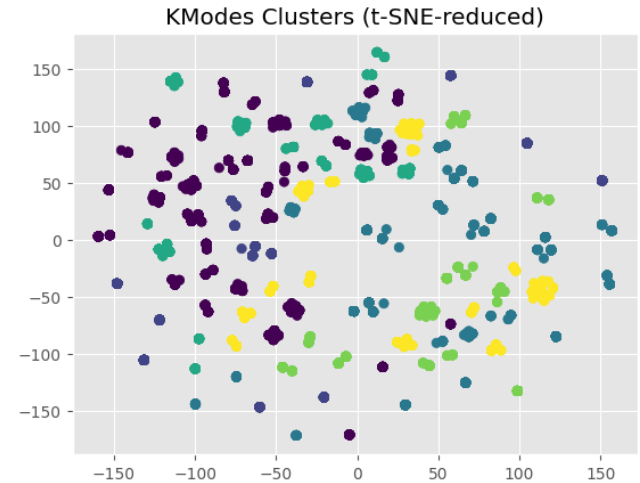
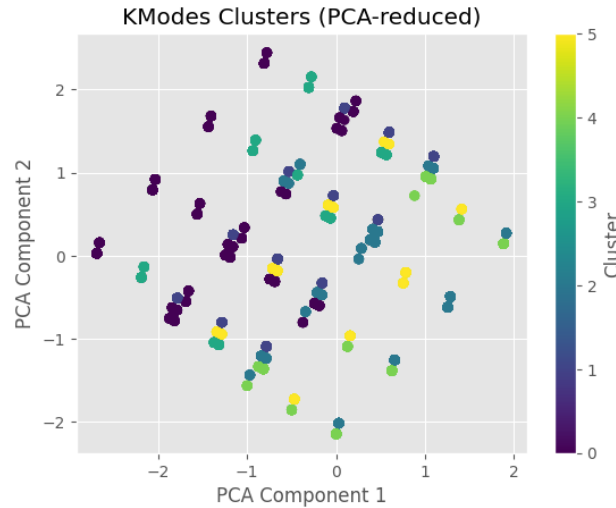
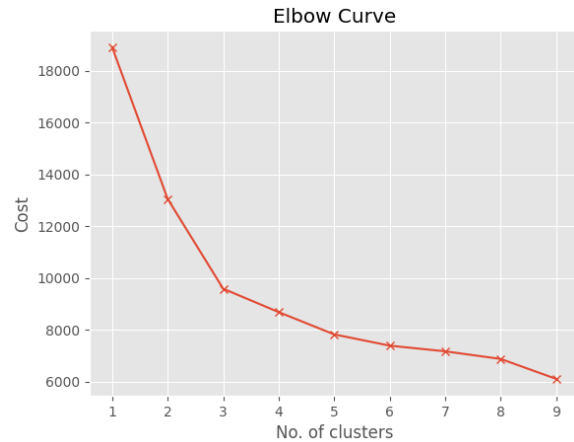
5. Results and Analysis



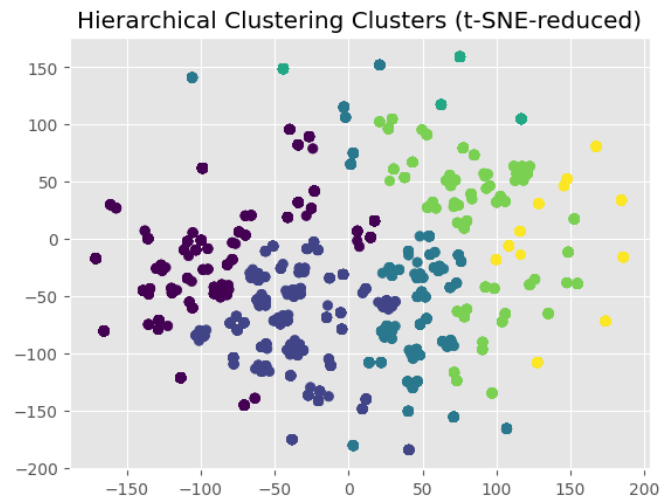
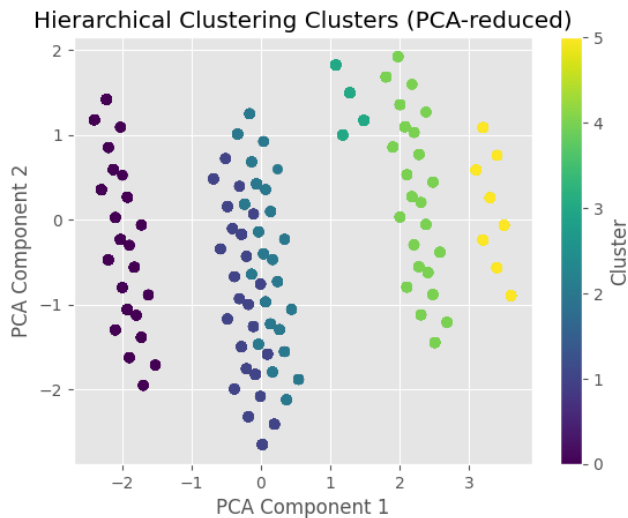
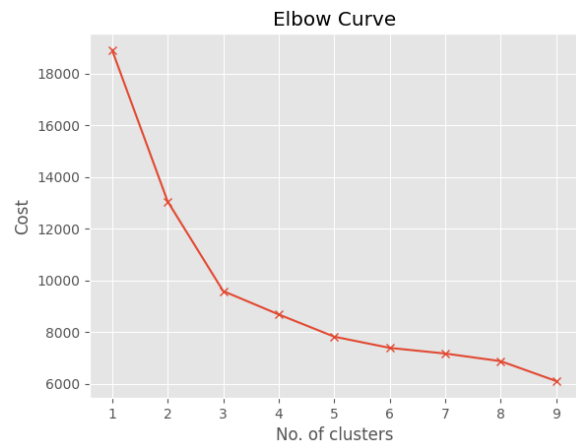
Epoch 1/10
221/221 — 2s 2ms/step - loss: 1.2029
Epoch 2/10
221/221 — 1s 2ms/step - loss: 1.0336
Epoch 3/10
221/221 — 1s 2ms/step - loss: 0.8308
Epoch 4/10
221/221 — 1s 3ms/step - loss: 0.6676
Epoch 5/10
221/221 — 1s 2ms/step - loss: 0.5849
Epoch 6/10
221/221 — 1s 3ms/step - loss: 0.5671
Epoch 7/10
221/221 — 1s 4ms/step - loss: 0.5538
Epoch 8/10
221/221 — 1s 6ms/step - loss: 0.5466
Epoch 9/10
221/221 — 1s 6ms/step - loss: 0.5413
Epoch 10/10
221/221 — 1s 4ms/step - loss: 0.5315
221/221 — 0s 2ms/step
mse:
6.118996053354586
relative_mse:
2.165359261913395



5. Results and Analysis



5. Results and Analysis



5. Results and Analysis

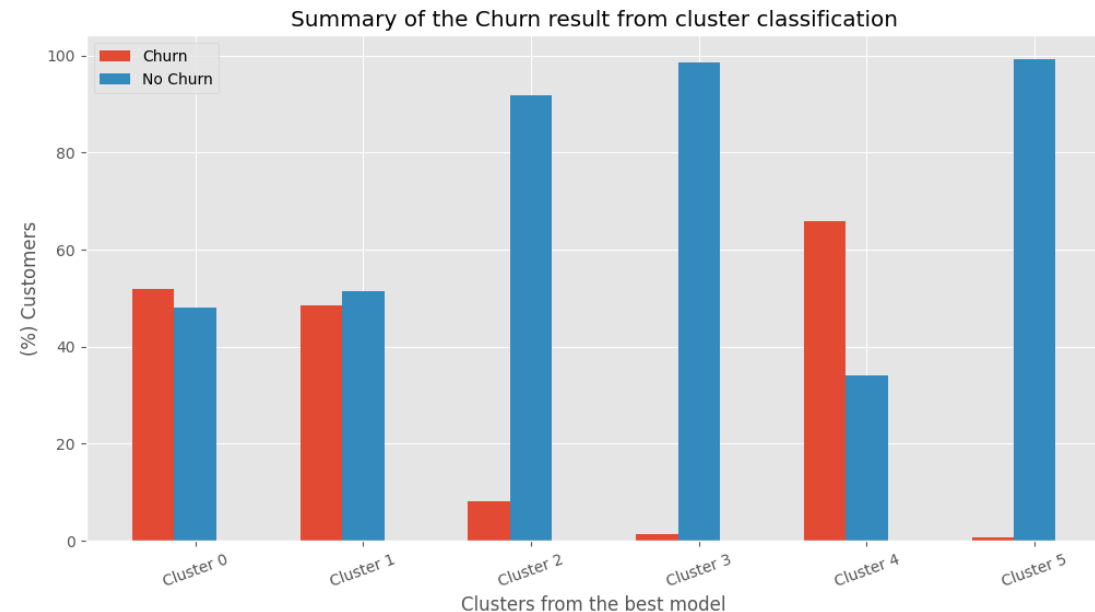
Visualize the best built model with the comparison of metrics for each model



Type of data	Model	Silhouette Score	Calinski-Harabasz Score	Davies-Bouldin Index
Numerical	K-Means	0.4	9595.31	0.87
Categorical	K-Modes	0.51	5275.08	0.94
Categorical	Hierarchical Clustering using Agglomerative Clustering	0.15	973.41	3.77

5. Results and Analysis, and Conclusion

Visualize the best built model with the comparison of customer churning for each cluster



Cluster	Internet Service	Online Security	Tech Support	Contract	Payment Method
0	Fiber optic	No	No	Month-to-month	Bank transfer (automatic)
1	DSL	No	No	Month-to-month	Electronic check
2	DSL	Yes	No	One year	Credit card (automatic)
3	DSL	Yes	Yes	Two year	Bank transfer (automatic)
4	Fiber optic	No	No	Month-to-month	Electronic check
5	No	No internet service	No internet service	Two year	Mailed check



Thank you